Mahivardhan Singh Panwar

TECHNICAL LEAD - ANDROID / GENERATIVE AI

+91 7742658282 | singhmahivardhan74@gmail.com | Bengaluru, Karnataka | LinkedIn

PROFESSIONAL SUMMARY

Results-driven Technical Lead with **10+ years** of experience architecting scalable **Android applications/SDK** and **2+ years** pioneering **Generative AI and Backend solutions**. Proven track record in designing high-performance SDKs using Clean Architecture, building enterprise-grade LLM integration platforms, and leading cross-functional teams across diverse domains including Telecom, US Healthcare, FinTech, and E-commerce. Expertise in transforming complex AI/ML concepts into production-ready microservices that drive business value.

CORE COMPETENCIES

Android Development

Kotlin | Jetpack Compose | Clean Architecture | MVVM | Coroutines/Flow | Dagger/Hilt

Android SDK and Libraries

Room Database | Retrofit | LiveData | RxJava | SDK Development | Mockito

Generative AI and LLM

Azure OpenAI | OpenAI API | Prompt Engineering | LLM Integration

Backend and DevOps

Python | FastAPI | REST APIs | Redis | Docker | Kubernetes

PROFESSIONAL EXPERIENCE

Technical Lead - Rakuten

May 2020 - Present
Bengaluru, India

Rakuten AI for Link - Voice AI SDK Android

View Project

Architected an intelligent AI agent SDK within Rakuten Link mobile app, leveraging LLM capabilities to deliver hyper-personalized recommendations and proactive issue resolution for millions of users.

- **Designed Voice AI SDK** providing chat interface for Japanese users with speech-to-text capabilities using Nuance DLGaaS (Dialog as a Service)
- Integrated with Nuance Mix for NLU, enabling natural language understanding across 40+ intent categories
- Implemented context-aware personalization engine analyzing user behavior patterns for tailored mobile experiences
- Built modular SDK architecture following Clean Architecture principles for easy integration into Link app

Tech: Kotlin, Nuance DLGaaS, Clean Architecture, MVVM, Coroutines, Hilt, Retrofit

My Rakuten Mobile - eCare SDK Android

View Project

Developed membership chatbot SDK enabling Rakuten Mobile subscribers to access billing statements, application status, and personalized support within the My Rakuten Mobile app.

- Implemented user experience scenarios with context-aware responses using Clean Architecture
- Integrated secure authentication flow for subscriber data access
- Built reusable SDK components for seamless integration across Rakuten Mobile apps

Tech: Kotlin, Retrofit, Coroutines, Clean Architecture, MVVM, Hilt, Room Database

CXD Personalization Engine Python Backend

Built a high-performance personalization microservice delivering real-time user profiling and intelligent recommendations for RMAA chatbot widget across Rakuten Mobile platform.

- **Architected backend with FastAPI** serving 100K+ daily decisions with less than 100ms latency
- Implemented **DoCoMo network detection** algorithm for intelligent carrier-based personalization
- Designed **rule evaluation engine** processing 15+ user profile enrichment facts for sophisticated targeting
- Built **Redis-based session management** with secure envelope passing between widget and chat backend
- Integrated with Profile Enrichment API for real-time user behavior analysis
- Implemented BigQuery analytics pipeline for comprehensive decision tracking and A/B testing

Tech: Python, FastAPI, Redis, BigQuery, Pydantic, Docker, Kubernetes

LLM Common Layer - Enterprise AI Platform Python Backend

Designed and developed a unified middleware platform enabling seamless LLM integration across multiple business units, eliminating code duplication and accelerating AI adoption.

- **Built multi-provider LLM abstraction layer** supporting Azure OpenAI and OpenAI APIs through a single unified interface
- Developed **self-service onboarding API** allowing teams to register API keys, models, business unit names, and custom prompts
- Implemented **Firestore-based configuration management** storing provider credentials, model mappings, and prompt templates
- **Reduced integration time from weeks to hours** - teams simply call the microservice with model name and BU identifier
- Created dynamic prompt injection system allowing teams to manage prompts via API without code changes
- Enabled secure multi-tenancy with isolated credential storage and audit logging per business unit

Tech: Python, FastAPI, Firestore, Azure OpenAI, OpenAI API, Docker

Senior Software Engineer - NextGen Healthcare

Apr 2018 - Apr 2020
Bengaluru, India

NextGen Mobile Solution (Entrada) Android

View Project

Healthcare application accelerating clinical documentation with AI-powered voice-to-text transcription for ambulatory and acute EHR systems.

- Designed application architecture enabling real-time audio-to-text conversion using Nvoq service

- Led code reviews and mentored team of 4 developers on Android best practices
- Coordinated with cross-functional teams (Design, Backend, QA) for product delivery

Tech: Android, Kotlin, RxJava, Retrofit, Room Database, MVVM

Senior Android Developer - MasterCard (PRI India Pvt Ltd)

May 2017 - Apr 2018
Vadodara, India

CitiPay - Mobile Wallet and NFC Payments (US) Android

Enterprise mobile wallet enabling cloud-based NFC payments for Citibank cardholders with secure tokenization.

- Architected payment flow with HCE (Host Card Emulation) for contactless transactions
- Implemented card provisioning and in-app checkout functionality
- Ensured PCI-DSS compliance in payment processing module

Tech: Android, HCE, NFC, Retrofit, Dagger, RxJava

Android Developer - Innovify

Oct 2013 - Apr 2017
Ahmedabad, India

OneWayCab Android

India's leading inter-city taxi booking platform with real-time driver tracking and seamless booking experience.

Tech: Android, Google Maps API, Retrofit, SQLite

Instantt Android

Local commerce app with Stripe wallet integration, location-based offers, and in-app payments.

Tech: Android, Stripe SDK, Google Places API, Retrofit

Bluetooth Remote Control Android

Cross-device control app for selfie capture and media playback via Bluetooth connectivity.

Tech: Android, Bluetooth API, Android Media APIs

KEY ACHIEVEMENTS

* Awarded prestigious **"E" certification** under Rakuten E-Rank Program (Jun 2022 - Dec 2022) - Top performer recognition
* Built LLM Common Layer reducing AI integration time from **weeks to hours** for 5+ business units
* Designed personalization engine processing **100K+ daily decisions** with sub-100ms latency

EDUCATION

Bachelor of Technology - Information Technology
Rajasthan Technical University, Kota